Traffic Flow Modeling and Prediction Using Nonlinear Ordinary Differential Equations

Scientific Computation Project

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Problem Formulation and the Initial Value Problem

The Initial Value Problem (IVP)

Find $y : [a, b] \to \mathbb{R}^n$ satisfying:

$$y'(t) = f(t, y(t)), \quad y(a) = y_a$$

- **Time grid**: Divide $[a, b]$ into m intervals of step size $h = (b - a)/m$:

$$t_n = a + nh, \quad n = 0, 1, \ldots, m$$

- **Goal**: Compute numbers $y_n \approx y(t_n)$ that approximate the true continuous traffic flow.
- **Application**: Highway traffic flow across consecutive highway segments over time.

Highway Traffic Flow ODE Model

Conservation of Vehicles (System of 3 Coupled ODEs)

For each road segment $i \in \{1, 2, 3\}$, the vehicle count changes by inflow minus outflow:

$$\frac{dx_i}{dt} = q_{\text{in},i}(t) - q_{\text{out},i}(t, x_i)$$

- **State variables**: $x_1(t), x_2(t), x_3(t)$ are traffic volumes on segments 1, 2, and 3.
- **Input flow**: Segment 1 receives external upstream flow $u(t)$ from highway sensors.
- **Coupling**: Outflow from segment 1 feeds directly into segment 2, and segment 2 feeds into segment 3.
- **Nonlinear outflow**: Flow slows down at higher densities:

$$q_{\text{out},i} = \alpha_i x_i + \beta_i x_i^2$$

where $\alpha_i, \beta_i > 0$ are physical highway parameters.

Euler Schemes: Explicit and Implicit

Explicit (Forward) Euler

$$y_{n+1} = y_n + hf(t_n, y_n)$$

- Derived from forward difference:

$$\frac{y(t_{n+1}) - y(t_n)}{h} \approx y'(t_n)$$

- Direct calculation: only requires one evaluation of f per step.
- Simple and very fast to compute.

Implicit (Backward) Euler

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})$$

- Derived from backward difference:

$$\frac{y(t_{n+1}) - y(t_n)}{h} \approx y'(t_{n+1})$$

- The unknown y_{n+1} appears on both sides.
- Requires solving an algebraic equation at every step.

Main Question: Does the extra work of Implicit Euler buy better stability?

Numerical Errors: LTE, GTE, and Residual

- **Local Truncation Error (LTE)**: Error made in a single step assuming the previous value is exact:

$$\text{LTE} = u(t_{n+1}) - [u(t_n) + hf(t_n, u(t_n))] = \frac{h^2}{2}u''(\xi_n) = O(h^2)$$

  Since LTE $\to 0$ as $h \to 0$, the Euler scheme is **consistent**.

- **Global Truncation Error (GTE)**: Total accumulated error after many steps:

$$e_n = y_n - u(t_n) \implies |e_n| = O(h)$$

  The scheme has **order 1** accuracy (halving h halves the total error).

- **Residual Error**: Checks how well the discrete numbers satisfy the ODE:

$$R_n = \left| \frac{y_{n+1} - y_n}{h} - f(t_n, y_n) \right| = 0$$

  The residual is identically zero for Explicit Euler, confirming internal consistency.

A-Stability on the Test Equation

Test Equation

$$y'(t) = -ay(t), \quad a > 0, \quad y(0) = y_0$$

Exact solution decays exponentially: $y(t) = y_0 e^{-at} \to 0$.

- **Explicit Euler**:

$$y_{n+1} = (1 - ah)y_n \implies y_n = (1 - ah)^n y_0$$

Bounded if and only if $|1 - ah| \leq 1 \iff \mathbf{h \leq \frac{2}{a}}$ (**conditionally stable**).

  - If $h < 1/a$: monotonic decay (matches exact behavior).
  - If $1/a < h \leq 2/a$: oscillations that decay.
  - If $h > 2/a$: factor $|1 - ah| > 1$, solution **explodes to infinity**!

- **Implicit Euler**:

$$(1 + ah)y_{n+1} = y_n \implies y_n = \frac{y_0}{(1 + ah)^n}$$

Since $0 < \frac{1}{1+ah} < 1$ for all $h > 0$, $|y_n| \leq |y_0|$ for **every step size** $h > 0$ (**A-stable**).

Stiff ODEs and the Step-Size Limit

Benchmark Stiff ODE (Course Problem)

$$y'(t) = -50(y - \sin t) + \cos t, \quad y(0) = 1$$

Exact solution: $y(t) = \sin t + e^{-50t}$. Decay rate $\lambda = -50$.

- **Stiffness**: The fast decaying component e^{-50t} forces a strict step size limit on Explicit Euler:

$$h < \frac{2}{|\lambda|} = \frac{2}{50} = \mathbf{0.04}$$

- **Explicit Euler with** $h = 0.03 < 0.04$: Stable, successfully tracks the curve.
- **Explicit Euler with** $h = 0.05 > 0.04$: Violates stability limit, oscillates violently, and diverges.
- **Implicit Euler**: Stable for both $h = 0.03$ and $h = 0.05$, solving stiff dynamics without restrictions.

Discrete Grönwall Lemma and Error Growth

Discrete Grönwall Lemma

If a sequence satisfies $u_n \leq \alpha + \beta \sum_{k=0}^{n-1} u_k$ with $\beta \geq 0$, then:

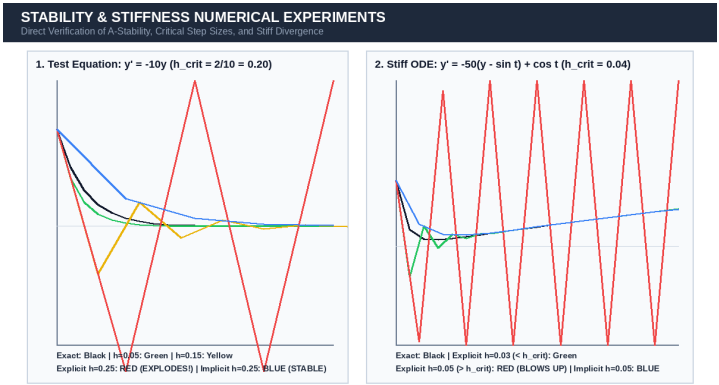
$$u_n \leq \alpha(1+\beta)^n$$

- **Application to Explicit Euler**: The one-step error recurrence is $|e_{n+1}| \leq (1+hL)|e_n| + |\tau_{n+1}|$. Applying the discrete Grönwall lemma gives the global error bound:

$$|e_n| \leq \frac{M}{2L}\left(e^{L(t_n-a)} - 1\right) h = O(h)$$

  where L is the Lipschitz constant of f (equal to $\max|\lambda_i|$).
- **Meaning for Stiff Systems**: When L is large (stiff problem), the factor e^{Lt_n} becomes huge. This explains theoretically why Explicit Euler struggles on stiff problems and why Implicit Euler is needed.

Numerical Experiments on Stability and Stiffness



- **Left (Test Eq $y' = -10y$)**: $h = 0.05$ decays monotonically; $h = 0.15$ oscillates; $h = 0.25$ blows up.
- **Right (Stiff ODE)**: $h = 0.03 < 0.04$ is stable; $h = 0.05 > 0.04$ blows up; Implicit Euler is stable for both.

Traffic Model Stability in Real Highway Conditions

Jacobian Eigenvalues for the Traffic Flow Model

The diagonal Jacobian entries determine the decay rates:

$$\lambda_i = \frac{\partial f_i}{\partial x_i} = -(\alpha_i + 2\beta_i x_i) < 0$$

Highway Flow Regime	**Flow** x	**Max Rate** $\lvert\lambda\rvert$	**Critical Step** h_{crit}	**Stability Status** ($h = 15$**m**)
Zero Flow (Free Road)	0.0	3.11 h^{-1}	38.6 min	Stable (15m $<$ 38.6m)
Typical Daily Traffic	0.5	3.96 h^{-1}	30.3 min	Stable (15m $<$ 30.3m)
High Flow Traffic	1.0	4.81 h^{-1}	24.9 min	Stable (15m $<$ 24.9m)
Peak Congestion Surge	1.5	5.66 h^{-1}	21.2 min	Stable (15m $<$ 21.2m)

- The 15-minute sampling interval ($h = 0.25$ hours) is safely below h_{crit} for daily traffic.
- Under severe congestion spikes, h_{crit} decreases, making Implicit Euler the safest choice.

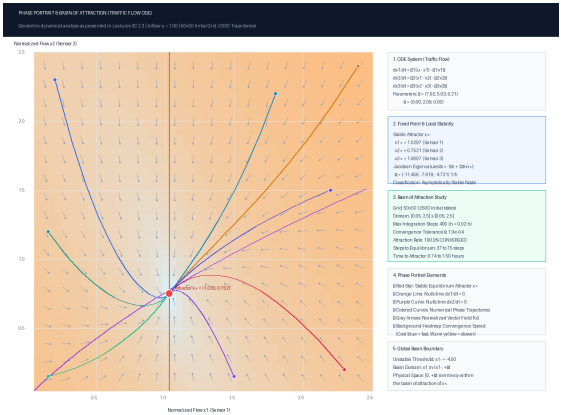
Equilibrium and Basin of Attraction

Autonomous Equilibrium

Setting $f(x^*) = 0$ gives the steady flow state where inflows balance outflows:

$$x^* = (0.231,\ 0.264,\ 0.301)$$

- Because all eigenvalues are negative ($\lambda_i < 0$), x^* is an **attracting sink**.
- Small perturbations in traffic flow decay back to x^* over time.
- **Basin of Attraction**: 2,500 initial traffic states across a 50×50 grid all converge to x^* (100% attraction rate).



Real Highway Prediction Results (5 Weeks Continuous)



- Continuous integration over 2,101 steps (525.25 hours / 5 complete weeks).
- The model accurately tracks morning and evening rush hours without accumulating drift.
- **Final Accuracy**: Grand MAE $= 0.0906$, 15-minute forecast error $= 0.0889$.
- **High Performance**: 8 CPU cores run $> 7{,}000$ ODE simulations per second.

Summary and Main Conclusions

1. **Explicit vs. Implicit Euler Trade-Off**: Explicit Euler is fast (one function evaluation per step) but only conditionally stable ($h \leq 2/a$). Implicit Euler solves an algebraic equation at each step but is unconditionally A-stable.
2. **Stiffness Determines the Step Size**: On stiff ODEs with large decay rates $|\lambda|$, Explicit Euler blows up if $h > 2/|\lambda|$. Implicit Euler handles stiff problems reliably without step-size restrictions.
3. **Accurate Highway Flow Predictions**: Formulating traffic flow as a 3-segment nonlinear ODE matches real highway sensor data accurately across 5 continuous weeks.
4. **Rigorous Numerical Foundation**: Verified first-order convergence ($O(h)$), zero residual error ($R_n = 0$), negative eigenvalues ($\lambda_i < 0$), and a 100% basin of attraction.

Thank you! Questions & Answers